

EPM213s - Electrical Measurements

Measuring internal inductance of moving iron

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1. Objective

This experiment aims to determine the internal AC impedance (|Zm||Z_m||Zm|) of a moving iron voltmeter and attempt to estimate its internal inductance (LmL_mLm) using measurements taken under alternating current (AC) conditions. The approach involves observing how the voltage across the instrument varies when resistors of known values are connected in series, using a stable AC voltage source at a fixed frequency.

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2. **Theoretical Background**

When a moving iron instrument with internal impedance Zm is connected in series with an external resistance Rs and an AC voltage source E_ac, the voltage V_ac measured across the instrument terminals can be related to these quantities. For a series circuit, if V_ac is the voltage across the meter (Zm) and Rs is in series with Zm, then the total impedance of the combination is Z_total = Zm + Rs (assuming Rs is purely resistive). The voltage across the meter is given by the voltage divider rule: V_ac = E_ac * |Zm / (Zm + Rs)| Rearranging this equation to match the user-provided method for analysis: E_ac / V_ac = |Zm + Rs| / |Zm| If we assume that for the purpose of this specific experimental analysis method, the relationship Y = Rs / |Zm| (where Y = E_ac/V_ac - 1) is used, similar to the DC case. This implies that |Zm| is treated as a constant base impedance and Rs adds to it in a way that the slope of Y vs Rs gives 1/|Zm|.

So, let Y = (E_ac / V_ac) - 1. Then, the user implies: Y = (1/|Zm|) * Rs

This equation is in the form of a straight line ($y = mx + c$) where Y is the dependent variable, Rs is the independent variable, the slope $m = 1/|Zm|$, and the intercept c should ideally be 0. By plotting Y against Rs, the meter's AC impedance $|Zm|$ can be determined from the reciprocal of the slope.

Once $|Zm|$ is found, and knowing the DC resistance Rm from Report 2, the internal inductance Lm can be calculated using the relationship for impedance in an AC circuit: $|Zm|^2 = Rm^2 + (\omega Lm)^2$ Where $\omega = 2\pi f$ is the angular frequency, and f is the frequency of the AC source. Therefore, $(\omega Lm)^2 = |Zm|^2 - Rm^2$ $Lm = \text{sqrt}(|Zm|^2 - Rm^2) / \omega$

3. Apparatus •

Moving Iron Instrument (Voltmeter)

AC Power Supply (Assumed $E_{ac} = 30V$, $f = 50Hz$)

Decade Resistance Box (for Rs)

Connecting Wires 4.

Experimental Data

The following readings were provided for an AC source voltage $E_{ac} = 30V$ at a frequency $f = 50Hz$:

Series Resistance (Rs) (Ohms)	Measured Voltage (V_ac) (Volts)
0	30.0
20	26.7
40	24.3
60	22.0
80	20.0

5. Calculations

Assuming the AC source voltage E_ac = 30V and frequency f = 50Hz. We calculate Y = (E_ac / V_ac - 1) for each set of readings:

Rs (Ohms)	V_ac (Volts)	E_ac/V_ac	Y = (E_ac/V_ac - 1)
0	30.0	1.000000	0.000000
20	26.7	1.123596	0.123596
40	24.3	1.234568	0.234568
60	22.0	1.363636	0.363636
80	20.0	1.500000	0.500000

Linear Regression

A linear regression was performed by plotting Y against Rs. * Independent variable (X-axis): Rs (Ohms) * Dependent variable (Y-axis): Y = (E_ac/V_ac - 1)

The results of the linear regression are: * Slope (m = 1/|Zm|): 0.006200 * Intercept (c): -0.003648 * R-squared value: 0.998699 (This indicates a very good linear fit)

Calculation of Meter Impedance |Zm|

Since the slope m = 1/|Zm|: |Zm| = 1 / slope |Zm| = 1 / 0.006200 **|Zm| = 161.29 Ohms**

The small non-zero intercept (-0.003648) is likely due to minor experimental errors or rounding.

Estimating L_m

From Report 2, $R_m = 179.69\,\Omega$. Using the formula:

$$(\omega L_m)^2 = |Z_m|^2 - R_m^2$$

$$|Z_m|^2 = (161.29)^2 = 26012.85$$
$$R_m^2 = (179.69)^2 = 32289.30$$

$$(\omega L_m)^2 = 26012.85 - 32289.30 = -6276.45$$

Since this result is **negative**, the square root yields an imaginary number—meaning L_m **cannot be calculated**, as impedance **must** be greater than resistance in an inductive AC circuit.

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6. Graphical Analysis

To visualize $|Z_m|$:

1. Plot $Y = (E_{\text{ac}}/V_{\text{ac}} - 1)$ on the vertical axis.
2. Plot R_s (Ohms) on the horizontal axis.
3. Draw the line of best fit through the data points.
4. The slope of this line equals $1/|Z_m|$.
5. Take the reciprocal to find $|Z_m|$.

Data for Plotting:

R_s (Ω)	$Y = (E_{ac}/V_{ac} - 1)$
0	0.000000
20	0.123596
40	0.234568
60	0.363636
80	0.500000

Expected Slope from Graph: ~0.006200

7. Conclusion

The AC impedance of the moving iron voltmeter, measured at 50 Hz, was determined to be **161.29 Ohms** based on experimental data and linear regression analysis.

However, an inconsistency arose during the attempt to estimate inductance. The calculated impedance $|Z_m|$ was found to be **lower than** the DC resistance R_m, which is **not physically possible** for a purely resistive-inductive system. This indicates either:

- Experimental error in voltage measurements
- Inaccuracies in source voltage levels
- Limitations in modeling the instrument as a simple RL circuit

As a result, a valid inductance value (L_m) could not be determined using the standard formula.